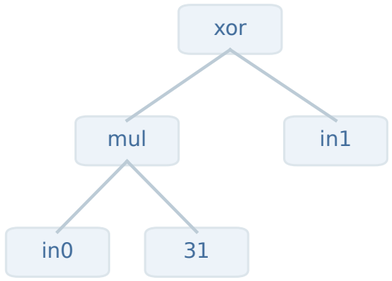
From RV32 computation to predicted LUT area

One fixed feature representation, followed by an equally weighted ensemble of 300 ExtraTrees.

01 INPUT

RV32 computation

```
(in0 × 31) XOR in1
```



32-bit integer semantics

02 FEATURES

Circuit representation

Static syntax + pre-mapping graphs

Original · normalized · operator graphs

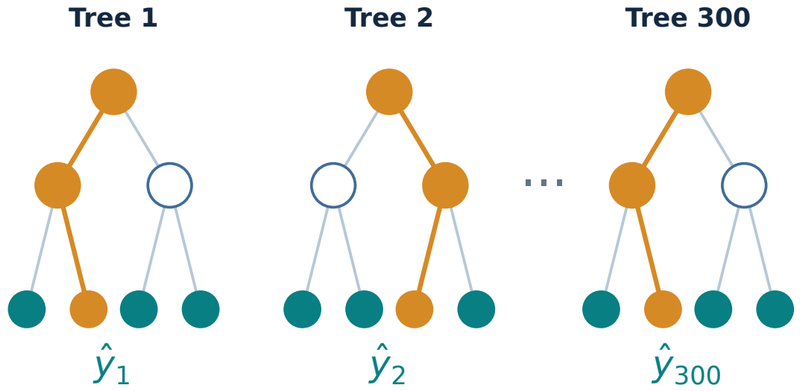
336
descriptors computed

217
fixed model inputs

03 FOREST

300 randomized decision trees

Each sees all training circuits and all 217 inputs.



Random split thresholds · No bootstrap sampling

04 OUTPUT

Mean

Equal weight

$$\hat{y} = \frac{1}{300}\sum_{t=1}^{300}\hat{y}_t$$

LUT4
predicted count

Area point estimate

OPTIONAL COVERAGE POLICY

Fixed arithmetic gate → **Base forest** or **Coverage forest** →

Return one selected prediction

Selection across models; averaging stays inside each forest.

Isolated original-RTL LUT4 area • uncalibrated points • no CPU speed, timing, power or uncertainty-bound claim